Lab 7: Bayesian Inference with MCMC

Instructor notes. Part 1: prep reading. Part 2: podium/circulating cheat sheet

How to prepare (for students)

- **Big picture.** Monday turned simulated annealing into the Metropolis algorithm, a way to draw samples from a posterior without ever computing Bayes' denominator. In this lab you run a chain by hand, learn to tell a trustworthy chain from an untrustworthy one, and then write your own sampler for a two-parameter model of chytrid fungus in frogs.
- **Review:** the Metropolis acceptance rule from Monday's lecture (and why a rejected step still gets recorded); the Beta-binomial posterior from Week 8; the logistic model and centering a predictor from Lab 6, Act 1.
- **Read:** Bolker 7.3 through 7.3.2 (MCMC, burn-in, convergence, printed p. 310-315).
- **R skills:** writing a function, `for` loops, `rnorm()`, `runif()`, storing results in a matrix row by row, `quantile()`.

Overview (instructor only)

Beat	Mode	Time
Opening	BOARD	~10 min
Act 1 (closed laptop)	paper	~55 min
Act 2 Part 1 + group plan check	paper	~30 min
Act 2 Part 2 (laptops)	laptop	rest of period; finish at home

Goal for the period: every group finishes Act 1, and every student leaves with an Act 2 plan their group has checked. Monday had no WOD, so the opening is short; skip it entirely if Monday's discussion (three germination chains) happened.

Group time for Act 2 (announce it; the handout doesn't say it). Act 2 has to work as an individual take-home, so it has no group language. When a group finishes Act 1, have them do Act 2's Part 1 (1a, 1b) on their own, then compare log posteriors and pseudocode as a group for ~10 min before opening laptops. Things to listen for: priors missing from 1a, a sign error turning it into an NLL, and pseudocode that records only accepted steps.

Data. Act 1 is one pond (2 of 20, weak Beta(2, 8) prior from nearby ponds, posterior Beta(4, 26)). Students add the log likelihood and log prior from a printed table to get each log posterior. Act 2 is a 12-pond survey that includes that pond (P11). The survey data are simulated (true $a = 2.2$, $b = -0.18$) and presented as field results. Keys: `labs/answer_keys/Lab7_Act1_Key.Rmd`, `Lab7_Act2_Key.Rmd`.

Where this came from. The Week 8 "if time" idea (reparameterize to remove a strong parameter correlation) appears here as Act 2's centering step. Bolker's 7.4 fitting checklist isn't used; it could open Wednesday or a later lab.

Part 1: Lab content

Opening (~10 min) [BOARD]

- One-line workflow; point at **FIT**.
- Metropolis on the board, in log form: propose $\theta^* = \theta + \text{jump}$; accept if $\log u < \log P(\theta^* \mid y) - \log P(\theta \mid y)$; record the current value either way.
- Three things to check in any chain (only if Monday's discussion was cut): burn-in (trend at the start), jumps too small (slow creep, near 100% accepted), jumps too big (flat stretches, few accepted). Runs from different starts should agree.
- Don't work an example with the lab's numbers. If students want one, use Monday's germination data (30 of 40).

Act 1 [paper]

1a answer (start 0.30):

Step	Proposal	Difference	Accept?	Record
1	0.22	+1.78	yes	0.22
2	0.31	−2.03	no ($\log u = -0.85$)	0.22
3	0.15	+1.00	yes	0.15
4	−0.02	$-\infty$	no	0.15
5	0.19	−0.49	yes ($\log u = -1.20$)	0.19
6	0.08	+0.58	yes	0.08

(Log posteriors: 0.30 is −3.02, 0.22 is −1.24, 0.31 is −3.27, 0.15 is −0.24, 0.19 is −0.73, 0.08 is −0.15.)

Sticking points: after the step 2 rejection, students start step 3 from 0.31 (ask "where is the chain right now?"); and some use only the log likelihood column and forget to add the log prior.

- **1b:** skipping rejections drops the repeats that weight the peak; chain too spread out.
- **1c:** constants cancel in the difference. That's Monday's $P(y)$ point again.

- **2a:** A good; B fine after ~150 steps of burn-in; C jumps too big (about 4% accepted, sticky); D jumps too small, two runs never meet.
- **2c:** credible 0.04 to 0.27 (mean 0.13), lopsided, above 0, and pulled up a little by the prior; Wald −0.03 to 0.23. Histogram should match Beta(4, 26).

Act 2 [paper → laptop]

- **2a check value:** -1.18 (a difference, so dropped constants don't matter). A value of +1.18 means they coded the negative log posterior.
- **2b, raw temperature:** ~20% accepted; two runs give e.g. -0.32 to -0.17 and -0.32 to -0.13. The runs disagree, which is the point.
- **2c, centered:** ~47% accepted; runs agree (-0.33 to -0.16, -0.33 to -0.15). Why: raw a is the log-odds at 0 °C, so a and b trade off (correlation about -0.97); independent jumps fall off the ridge. Same as the Lab 6 beetle dose and Monday's myxo banana.
- **2d:** Wald -0.32 to -0.15; agrees with the credible interval to ~0.01.
- **Common bugs:** current log posterior not updated after an acceptance (acceptance rate drifts toward 100%); storing only on acceptance; `quantile()` over the whole matrix; same seed for both runs.
- **Reflection target:** many data and no boundary, so the intervals agree; Act 1 had few data near $p = 0$, so they didn't.

PART 2: CIRCULATING CHEAT SHEET

Terse on purpose. Detail: Part 1 and the keys.

Opening (~10 min) [BOARD]

- Workflow; point at **FIT**
- Accept if $\log u < \text{lp(prop.)} - \text{lp(curr.)}$; **record either way**
- If Monday's discussion was cut: burn-in, too small, too big, runs agree
- Examples: germination (30 of 40) only, never the lab's numbers

Act 1 (~55 min) [paper]

- Records: 0.22, 0.22, 0.15, 0.15, 0.19, 0.08
- lp = log lik. + **log prior** (both from the table)
- Step 3 starts from **0.22**, not 0.31
- Step 4 (-0.02): reject, posterior 0
- 1c: constants cancel
- A good; B burn ~150; C too big; D too small, runs apart
- Credible 0.04-0.27 vs. Wald -0.03-0.23; Beta(4, 26)

Act 2 (rest of period) [paper → laptop]

- **After Act 1:** Part 1 alone, then ~10 min group compare before laptops
- Listen for: no priors; NLL sign; records only acceptances
- 2a check: -1.18
- Raw: ~20% accepted, runs disagree
- Centered: ~47%, runs agree, b about -0.33 to -0.16
- Wald -0.32 to -0.15
- Bugs: lp not updated; store only on accept; same seed twice
- Unfinished? Finish at home